Inferential Analysis of Tooth Growth

Kimberly Bormann

2025-04-06

Overview

This document explores the effects of vitamin C on tooth growth in guinea pigs. We will compare vitamin C delivered through orange juice (OJ) or ascorbic acid (VC) at dosages of 0.5, 1, and 2 milligrams in relation to overall tooth growth.

Data Summary

We begin by loading the data and inspecting it.

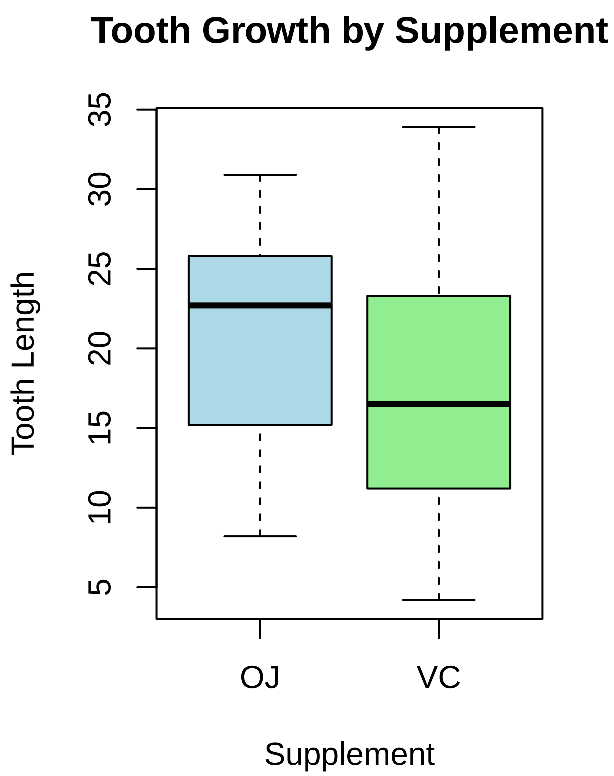
```r
#load package and data
library(datasets)
data("ToothGrowth")
#explore data
summary(ToothGrowth)
```

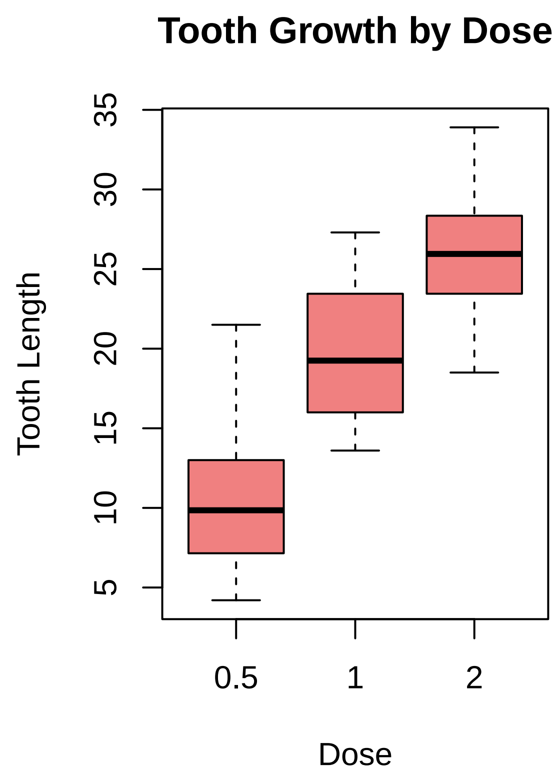
```
##       len        supp         dose      
##  Min.   : 4.20   OJ:30   Min.   :0.500  
##  1st Qu.:13.07   VC:30   1st Qu.:0.500  
##  Median :19.25           Median :1.000  
##  Mean   :18.81           Mean   :1.167  
##  3rd Qu.:25.27           3rd Qu.:2.000  
##  Max.   :33.90           Max.   :2.000  
```

We can see that there are three variables: tooth growth, type of Vit C supplement, and dose of supplement. We can also see that the tooth growth has a wide range from 4.20 to 33.9 and the dose options are 0.5, 1.0, and 2.0.

To better understand the effects of these supplements, we will investigate further with some exploratory graphs.

```r
#set up the plotting area
par(mfrow = c(1, 2))
#boxplot of tooth growth by supplement type
boxplot(len ~ supp, data = ToothGrowth,
        main = "Tooth Growth by Supplement",
        xlab = "Supplement",
        ylab = "Tooth Length",
        col = c("lightblue", "lightgreen"))
#boxplot of tooth growth by dose
boxplot(len ~ dose, data = ToothGrowth,
        main = "Tooth Growth by Dose",
        xlab = "Dose",
        ylab = "Tooth Length",
        col = "lightcoral")
```





Tooth Growth by Supplement: The boxplot comparing tooth growth by supplement type indicates that the median tooth growth is higher for the OJ group compared to the VC group. However, the interquartile ranges for both groups largely overlap, suggesting that the spread of the central 50% of the data is similar for OJ and VC. This overlap implies that while there is a difference in the central tendency, the overall distribution is not drastically different.

```
mean(ToothGrowth$len[ToothGrowth$supp == "OJ"]) - mean(ToothGrowth$len[ToothGrowth$supp == "VC"])
```

The difference in tooth growth between groups is 3.7 units. Considering the broad range of tooth growth observed, this difference may be modest.

Tooth Growth by Dose: In contrast, the boxplots for tooth growth by dose display a clear upward trend: as the dose increases, the median tooth growth increases, and the entire interquartile range shifts upward. This separation between the dose groups suggests that the dose has a pronounced effect on tooth growth. The increasing medians and non-overlapping interquartile ranges across doses provide stronger evidence that the supplement dose is a significant factor in determining tooth growth.

Tests of Statistical Significance

Our exploratory analysis with boxplots provided initial evidence of differences in tooth growth by both supplement type and dose. However, we cannot rely solely on descriptive statistics to draw definitive conclusions. Therefore, we will perform two t-tests to assess whether these differences are significant.

T-test for Tooth growth between Supplements

We begin by performing a two-sided t-test to compare the mean tooth growths between the OJ and VC groups. This conservative approach is used because we do not have a strong prior expectation that one group's mean would be higher than the other's.

The box plot suggests that we can assume equal variance for this test since the distribution of data appears similar. We recall in our summary that both groups have a sample size of 30. Since they are the same we will assume equal variance.

The null hypothesis is that the mean of each group is equal. The alternative hypothesis is that the mean of each group is not equal.

```r
t_test_supp <- t.test(len ~ supp, data = ToothGrowth, var.equal = TRUE )
t_test_supp
```

```
##
##  Two Sample t-test
##
## data:  len by supp
## t = 1.9153, df = 58, p-value = 0.06039
## alternative hypothesis: true difference in means between group OJ and group VC is not equal to 0
## 95 percent confidence interval:
##  -0.1670064  7.5670064
## sample estimates:
## mean in group OJ mean in group VC
##         20.66333         16.96333
```

The 95% confidence interval is about [-0.167, 7.567]. Since 0 is a value in the confidence interval it supports the conclusion that the difference in mean might be zero. Since the p-value is about 0.060, which is above the 0.05 significant level, we do not have enough evidence to support that the means differ.

Therefore, we accept the null hypothesis that there is no statistically significant difference between the OJ group and VC group. **This evidence suggests that tooth growth does not differ significantly between supplement types.**

T-test for Tooth growth between Dose

We will now perform multiple comparisons for tooth growth between dose pairs with separate t-tests for each pair (0.5 vs. 1.0, 0.5 vs. 2.0, and 1.0 vs. 2.0). We will adjust for multiple comparisons using the Bonferroni method, which divides our significance level by the number of test making it quite conservative.

The null hypothesis is that the mean of each group is equal. The alternative hypothesis is that the mean of each group is not equal.

```r
pairwise.t.test(ToothGrowth$len, ToothGrowth$dose, p.adjust.method = "bonferroni", var.equal = TRUE)
```

```
##
##  Pairwise comparisons using t tests with pooled SD
##
## data:  ToothGrowth$len and ToothGrowth$dose
##
##   0.5     1
## 1 2.0e-08 -
## 2 4.4e-16 4.3e-05
##
## P value adjustment method: bonferroni
```

The p values for comparison between dose 0.5 and dose 1, dose 0.5 and dose 2, and 1 and dose 2 are 2.0×10^{-8}, 4.4×10^{-16}, and 4.3×10^{-5} respectively.

All these p values are extremely small and well below the significance level of 0.05, indicating that the difference in mean tooth growth between doses is highly significant.**These results strongly suggest that tooth growth differs significantly between all pairs of dose levels.**